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February 27, 1974
SUPERSEDING
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March 20, 1967

FEDERAL SPECIFICATION

LATH, AND BOARD PRODUCTS, GYPSUM

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers gypsum lath, gypsum sheathing boards, gypsum wallboards, gypsum backing boards, gypsum form boards, gypsum veneer plaster base, water-resistant gypsum backing boards, and fire barrier gypsum boards.

1.2 Classification. Gypsum boards covered by this specification shall be of the following types, grades, classes, forms, and styles as specified (see 6.2).

Type I - Lath.
 Grade R - Regular Core.
 Grade X - Fire Retardant Core.
  Class I - Plain face.
    Form a - Plain back.
    Form b - Perforated.
    Form c - Foil back.
     Style 1 - Square edge.
     Style 5 - Round edge.

Type II - Sheathing.
 Grade R - Regular Core.
 Grade W - Water Resistant Treated Core.
 Grade X - Fire Retardent Core.
  Class 2 - Water Resistant Surfaces.
    Form a - Plain back.
     Style 1 - Square edge.
     Style 2 - V-Tongue and groove edge.

Type III - Wallboard.
 Grade R - Regular Core.
 Grade X - Fire-Retardant Core.
  Class 1 - Plain face.
  Class 3 - Predecorated face.
    Form a - Plain back.
    Form c - Foil back.
     Style 1 - Square edge.
     Style 3 - Tapered or Recessed edge.
     Style 4 - Featured Joint edge.
     Style 6 - Tapered having featured edge.

Type IV - Backing board.
 Grade R - Regular Core.
 Grade X - Fire Retardant Core.
  Class 1 - Plain face.
    Form a - Plain back.
    Form c - Foil back.
     Style 1 - Square edge.
     Style 2 - V-Tongue and groove edge.

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Type V - Formboard.
Grade R - Regular Core.
Class 4 - Fungus-Resistant Surface.
Form a - Plain back.
Style 1 - Square edge.

Type VI - Veneer Plaster Base.
Grade R - Regular Core.
Grade X - Fire Retardent Core.
Class 1 - Plain face.
Form a - Plain back.
Form c - Foil back.
Style 1 - Square edge.
Style 3 - Taper or Recess edge.
Style 6 - Tapered having featured edge.

Type VII - Water-Resistant gypsum backing board.
Grade R - Regular Core.
Grade W - Water Resistant Treated Core.
Grade X - Fire Resistant Treated Core.
Class 2 - Water Resistant Surface.
Form a - Plain back.
Style 1 - Square edge.
Style 3 - Tapered or Recess edge.

Type VIII - Fire barrier gypsum board.
Grade X - Fire Retardant Core.
Class 2 - Water Resistant Surface.
Class 4 - Fungus Resistant Surface.
Form a - Plain back.
Style 1 - Square edge.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-W-3448 - Wallboard: Packaging of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Sepcifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

C 36 - Specification for Gypsum Wallboard.
C 37 - Specification for Gypsum Lath.
C 79 - Specification for Gypsum Sheathing Board.
C 318 - Specification for Gypsum Formboard.
C 442 - Specification for Gypsum Backing Board.
C 473 - Method for Testing of Gypsum Board Products and Gypsum Partition Tile on Block.
C 588 - Specification for Gypsum Base for Veneer Plastics.
C 630 - Specification for Water-Resistant Gypsum Backing Board.
C 707 - Specification for Fire Barrier Gypsum Board Membrane.
E 84 - Test for Surface Burning Characteristics of Building Materials.
E 119 - Fire Tests of Building Construction and Materials.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. The panels shall consist of a gypsum core, with or without fiber reinforcements, and shall be surfaced on both faces and longitudinal edges with a tightly bonded material. The core of the boards shall be predominately gypsum. Additives shall not exceed 15 percent by weight. The surface material of the boards shall be paper on similar fiberous material. Plastic facings or coatings shall be permitted on predecorated (class 3) boards as specified (see 6.2).

3.2 Types. Unless otherwise specified herein, gypsum boards shall meet the requirements of the applicable ASTM Standard specification listed in table I for the type board required. Quality assurance, packing and marking requirements shall be as specified in this specification (see 6.2). Gypsum boards shall meet the required performance characteristics, board sizes, and dimensional tolerances specified in tables II, III, IV and V when tested in accordance with 4.4. Flame spread ratings shall be as specified in the applicable ASTM Standard when tested in accordance with 4.4.4.

TABLE I. ASTM Standards

Type Gypsum Board	ASTM Designation
Type I - Gypsum Lath.	ASTM C 37
Type II - Gypsum Sheathing Board.	ASTM C 79
Type III - Gypsum Wallboard.	ASTM C 36
Type IV - Gypsum Backing Board.	ASTM C 442
Type V - Gypsum Formboard.	ASTM C 318
Type VI - Gypsum Veneer Plaster Base.	ASTM C 588
Type VII - Gypsum Water Resistant Backing Board.	ASTM C 630
Type VIII - Gypsum Fire Barrier Board.	ASTM C 707

3.3 Sizes. The gypsum boards shall be of the nominal dimensions shown in table II as specified, (see 6.2).

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TABLE II. Board sizes

Type	Thickness	Width	Length
	Inch	Inches	Inches
I	3/8, 1/2	16, 16-1/5 or 24	48 to 144
II	1/2	24 or 48	96 or 108
III	1/4, 3/8, 1/2, 5/8 or 3/4	16, 24, 30, 32 or 48	1/ 48 to 192
IV	1/4, 3/8, 1/2, 5/8 or 1	16, 24, 32 or 48	1/ 48 to 192
V	1/2	32	48 to 120
VI	3/8, 1/2, or 5/8	48	48 to 192
VII	1/2 or 5/8	48	48 to 192
VIII	1/2	24 or 48	96

1/ Maximum allowable lengths of 1/4 inch boards shall be limited to 144 inches.

3.3.1 Dimensional Tolerance. The maximum variation in board sizes shall not exceed the tolerances specified in table III.

TABLE III. Dimensional tolerance

Type	Thickness		Width	Length
	Local variation	Average variation		
	Inch	Inch	Inch	Inch
I	± 1/16	± 1/32	+ 1/8 - 3/16	+ 1/8 - 1/4
II, IV	± 1/16	± 1/32	+ 1/8	+ 1/4
III, VI, VII	± 1/32	± 1/64	± 0 - 3/32	± 1/4
V	± 1/32	± 1/64	+ 1/8 - 1/4	± 1/4
VIII	± 1/32	± 1/64	± 1/8	± 1/8

3.4 Weight limitations. The weight of the gypsum boards per 1,000 square feet shall conform to the requirements of the IV.

TABLE IV. Weight limitations

Type	Thickness	Minimum weight
	Inch	lb/1,000 sq. ft.
III and IV	1/4	900
I, III, IV, VI	3/8	1300
III, IV, and VIII	1/2	1700
I, II, V, VI, VII	1/2	1800
III, IV, VII	5/8	2100
VI	5/8	2250
III	3/4	2800
IV	1	3400

3.5 Flexural strength. When tested in accordance with 4.4.5 the breaking load shall be not less than the loads specified in table V.

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TABLE V. Flexural strength loads

Type	Form	Thickness Inch	Load across fiber Pounds	Load parallel to fiber Pounds
I	a, c	3/8	60	27
	a, c	1/2	100	40
	b	3/8	60	20
	b	1/2	100	30
II	a	1/2	110	50
III	a, c	1/4	50	24
	a, c	3/8	80	34
	a, c	1/2	110	50
	a, c	5/8	150	60
	a	3/4	185	72
IV	a, c	1/4	50	24
	a, c	3/8	70	31
	a, c	1/2	100	40
	a, c	5/8	145	55
	a, c	1	225	90
V	a	1/2	160	60
VI	a, c	3/8	80	34
	a, c	1/2	105	45
	a, c	5/8	145	55
VII	a	1/2	110	50
	a	5/8	150	60
VIII	a	1/2	110	50

3.6 Grades.

3.6.1 Grade R, regular core. Regular core lath and gypsum boards without special core addative to improve water and fire resistance.

3.6.2 Grade W, water resistant treated core. Grade W water resistant treated core sheathing boards shall be treated to prevent the absorption of more than 10 percent of water by weight, after 2-hour immersion, and Grade W - water resistant backing boards shall be treated to prevent the absorption of more than 5 percent of water by weight after 2-hours immersion, when tested in accordance with 4.4.2.

3.6.3 Grade X, fire retardant core. Fire retardant core lath and boards shall be manufactured to provide at least the equivalent of either a 1-hour fire resistance rating for 5/8 inch thick or a 3/4 hour fire resistance rating for 1/2-inch thick lath, or type panel specified (see 6.2), when applied in single-layer nailed application on each face of load-bearing 2 x 4 nominal size wood-framing members, spaced 16 inches on center, and tested in accordance with 4.4.3.

3.7 Classes.

3.7.1 Class 1, plain face. Plain face lath or boards shall be true and free of imperfections which may impair the serviceability of the boards.

3.7.1.1 Lath, or veneer plaster base surface. The plain face of faces and the longitudinal edges shall readily receive and retain gypsum plaster.

3.7.1.2 Wallboard surface. The surface of the face intended for exposure after erection shall readily receive and retain decoration. The back face shall be faced with a liner paper.

3.7.1.3 Backing board surface. Both faces and longitudinal edges of the backing boards shall be surfaced with a liner paper.

3.7.2 Class 2 - water resistant surface. The faces and longitudinal edges shall be covered with water resistant paper. The surfaces shall be free of openings to the core.

3.7.3 Class 3 - predecorated surface. The surface of the face intended for exposure after erection shall be decorated in such a manner that no additional decoration is necessary. The decorated surface shall be true and free of imperfections.

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3.7.4 Class 4 - fungus-resistant surface. The surface intended for exposure and longitudinal edges shall be covered with a calendered manila or grey paper which has been treated to resist fungus growth. The back face shall be surfaced with liner paper. Certification shall be furnished in accordance with 4.4.7.

3.8 Forms.

3.8.1 Form a - plain back. Plain back boards shall have no surface which will prevent the normal passage of heat or vapor and shall not be perforated.

3.8.2 Form b, perforated. Perforated boards shall have nominal 3/4-inch diameter perforations, with at least one perforation for each 16 square inches of surface face, and shall not be insulated.

3.8.3 Form c, foil back. Foil back boards shall have pure, bright, finished, aluminum foil tightly bonded to the face intended to be concealed after erection.

3.8.3.1 Vapor permeance. The vapor permeance of foil back boards shall not exceed 0.30 perm for the condition of 50 percent relative humidity on the exposed face and 0 percent humidity on the foil back face.

3.9 Styles. The short (end) edges of the boards shall be square, butt edges.

3.9.1 Style 1, square edged. The long edges shall be square.

3.9.2 Style 2, V-tongue and groove edged. One long edge shall have a V-tongue, and the other edge shall have a matched groove.

3.9.3 Style 3, taper or recess edged. The long edge shall be tapered or recessed to a thickness at least 0.015-inch less than the thickness of the interior field of the board, but not more than 0.075-inch less than the thickness of the interior field of the board, when tested in accordance with 4.4.6.

3.9.4 Style 4, featured joint edged. The long edges shall be matching edges intended to form an exposed joint.

3.9.5 Style 5, round edge. The long edges shall be uniformly rounded.

3.9.6 Style 6, tapered, having featured edges. The long edges shall be uniformly tapered with featured edges.

3.10 Marking. Boards shall be marked with the manufacturer's name or brand. Boards, except grade X, having a face intended for exposure after erection, shall be marked on the surface intended to be concealed. Grade X boards shall be marked with the manufacturer's name or brand and an appropriate fire-resistance rating designation.

3.11 Workmanship. Boards shall be whole and clean. All edges shall be straight and solid. Corners shall be square, with a permissible variation of 1/8-inch in the full width of the board except for lath which will be 1/4-inch.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all panels of the same type, grade, class, form, style and size offered for delivery at the same time.

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4.2.2 Visual and dimensional examination. Sampling for this examination shall be in accordance with level S-4 in MIL-STD-105. The acceptable quality level (AQL) for defects listed in table VI shall be 2.5 percent defective for major defects, and 6.5 percent defective for minor defects.

4.2.3 Sampling for tests. Sampling for tests shall be in accordance with inspection level S-4 in MIL-STD-105, for all boards except grade X - fire-retardant core boards. Grade X boards shall be tested in accordance with 4.2.3.1.

4.2.3.1 Sampling for fire-retardant rating test. The sample shall consist of enough panels to cover both sides of wood-bearing members in a single layer for an area of not less than 100 square feet, with no dimension of the area less than 9 feet. Failure of the sample to attain the specified rating shall be cause for rejection of the lot.

4.3 Examination.

4.3.1 End item. Samples selected in accordance with 4.2.2 shall be examined for defects listed in table VI.

TABLE VI. Classification of defects

Defects	Major	Minor
Thickness, width or length not as specified.....	X	
Material not as specified.....	X	
Weight not as specified.....	X	
Surfaces not as specified.....	X	
Class 2 board not covered with water resistant surface or contains openings to core.....	X	
Perforations not as specified (Form b).....	X	
Marking omitted or not as specified.....	X	
Edges not as specified.....	X	
Damage or defects not affecting function or serviceability.		X

4.3.2 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing, and marking requirements comply with section 5. Defects shall be scored in accordance with table VII. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 6.5 defects per hundred units.

TABLE VII. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strappings, inadequate stapling, not sealed as specified for level A. Bulging or distortion of container.
Contents	Number per container is more or less than required. Net weight excedes the requirements of the applicable box specification.

4.3.3 Examination. The packing and marking of the lath, sheathing, and wallboards shall be examined and tested to determine compliance with MIL-W-3448.

4.4 Test methods.

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4.4.1 Test specimens. Samples selected for testing shall be marked, scored on both faces, and a 1-foot piece broken off either end and discarded. The specimen shall be used for determining taper or recess thickness. After use in the taper or recess thickness tests the specimen shall be retained as a source of specimens for flexual strength test Vapor permeance test, and water-resistant rating test, as applicable.

4.4.2 Water resistance. The moisture absorption of grade W panels shall be determined in accordance with the method described in sections 19, 20, and 21 of ASTM C 473, (see 3.6.2).

4.4.3 Fire-resistance rating (Grade X). The fire resistance rating of grade X panels shall be determined in accordance with the tests of bearing walls and partitions described in ASTM E 119, or any equivalent fire endurance test method certified by the manufacturer (see 3.6.3).

4.4.4 Flame spread ratings. The flame spread ratings of each type panel shall be determined in accordance with ASTM E 84 (see 3.2).

4.4.5 Flexural strength. Flexural strength of specimens shall be determined in accordance with the methods described in sections 5, 6, and 7 of ASTM C 473 (see 3.5).

4.4.6 Taper or recess thickness test. The taper or recess thickness of style 3 panels shall be determined in accordance with the methods described in section 16, 17 and 18 of ASTM C 473 (see 3.9.3).

4.4.7 Fungus-resistance certification. There is no short-term test which will readily determine conformance to the fungus-resistance requirements for class 4 board. In lieu of a test, the manufacturer shall furnish a certification that the board has been surface treated to resist the growth of fungus (see 3.7.4).

5. PREPARATION FOR DELIVERY

5.1 Packing. The lath, sheathing and wallboards shall be packed and marked in accordance with MIL-W-3448. Level of packing shall be level A or level C in accordance with MIL-W-3448 as specified (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use:

Type I lath is intended for use in interior sidewalls, partitions, and ceilings as a base for gypsum plaster. The lath is not suitable as a base for portland cement or lime plaster, and the foil side of insulating lath is not adapted to receive or retain plaster.

Type II sheathing is intended for use on exterior sidewalls and is normally a base for a finish covering such as siding.

Type III wallboard is intended for use as a covering for interior sidewalls, partitions or ceilings, with decoration such as paint or wallpaper, or without further decoration when predecorated.

Type IV backing board is intended for use as a base in multiple layer gypsum board assemblies either supported on studding or as unsupported partition. By laminating wallboard over backing board, nails can be effectively concealed.

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Type V formboard is intended for use as permanent forms for gypsum concrete roof decking. The interior boards are not recommended for use where they are exposed to excessive moisture. When interior boards are exposed in roof spaces, adequate ventilation to prevent moisture accumulation is essential. Gypsum board 3/8-inch thick should not be used where support spacing exceeds 16 inch center, and 1/2-inch boards should not be used where support spacing exceeds 24-inch centers.

Type VI gypsum base for veneer plasters are designed to be used as a base for application of veneer plaster with the same limitations as specified for type I lath above.

Type VII water-resistant gypsum backing board is designed for use in bath and shower areas as a base for the application of ceramic or plastic tile.

Type VIII fire barrier gypsum board membrane is designed to be used as a fire barrier in roof, exterior wall, and floor-ceiling assemblies.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, grade, class, form, and style required (see 1.2.1).
(c) Dimensions required (see 3.2).
(d) Selection of applicable level of packing and marking required (see 5.1).

MILITARY INTEREST

Custodians:

Army - ME
Navy - MD
Air Force - 84

Review activities:

Army - ME
Navy - SA
Other - NSA

MILITARY COORDINATION ACTIVITY:

Army - ME

Preparing activity:

GSA-FSS

Civil Agency Coordinating Activity:

COMMERCE-NBS
GSA-FSS
HEW-FEC
VA-OMS

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